Extended reading — Week 11: Contextual Bandits and Off-Policy Evaluation

October 6, 2026

Five papers to read this week

The five papers below cover the spine of the bandit + OPE literature you will lean on for synthetic insurance / ATS work: the original UCB1 analysis, the LinUCB news-recommendation paper that founded contextual bandits as a deployable technique, the LinTS analysis that turned posterior sampling into a defensible default, the Doubly Robust estimator (the only OPE tool you should reach for first), and BanditNet (counterfactual risk minimisation at deep-net scale).

1. Auer, Cesa-Bianchi, Fischer — *Finite-time Analysis of the Multiarmed Bandit Problem* (Machine Learning, 2002)

The paper that turned the 30-year-old Lai–Robbins asymptotic into a finite-time, instance-dependent regret bound. UCB1 is presented in two pages; the rest is a careful Hoeffding-based proof and an extension to the variance-aware UCB1-tuned.

Why it matters this week. Every bandit paper since cites this one. The proof technique — "either a bad concentration event happened, or $N_a \geq 8 \log T / \Delta_a^2$" — is the canonical bandit proof template. You will see it repeated for LinUCB, KL-UCB, and MOSS.

> "At each play, UCB1 maintains a confidence bound for the mean reward of each arm. The policy chooses the arm with the largest upper confidence bound. We show that the expected regret of this policy after any number n of plays is at most $O(\log n)$."

Commentary. The single sentence above is the entire premise of OFU (optimism in the face of uncertainty). The $O(\log n)$ rate is tight up to constants; the instance-dependent $1/\Delta_a$ scaling is the bit you must keep in mind when reasoning about *when* bandit algorithms are slow — small Δ_a means hard problem, and no algorithm can escape this.

2. Li, Chu, Langford, Schapire — *A Contextual-Bandit Approach to Personalized News Article Recommendation* (WWW 2010, arXiv:1003.0146)

The paper that put contextual bandits on the industry map. Two algorithms (LinUCB-disjoint, LinUCB-hybrid), deployed on Yahoo! Today Module, evaluated by a clever *replay* estimator that turns logged uniform exposures into unbiased offline evaluations.

Why it matters this week. The disjoint-arms LinUCB you implement in `project/agents.py` is the LinUCB from this paper, with the tightened confidence radius from Abbasi-Yadkori et al. 2011. The hybrid variant (a shared feature across arms) is what you would deploy in synthetic insurance when most brokers have few observations.

> "If we constrain the design of the experiment so that articles are chosen uniformly at random in the recorded events, then replay yields an unbiased offline evaluation."

Commentary. This is the simplest OPE estimator possible — IPS with a uniform behaviour policy collapses to "keep matches, drop misses". It works because uniform exposure gives perfect overlap. Production loggers are almost never uniform; you need the IPS / DR machinery from this week.

3. Agrawal & Goyal — *Thompson Sampling for Contextual Bandits with Linear Payoffs* (ICML 2013, arXiv:1209.3352)

The frequentist regret analysis of LinTS. Earlier work (Chapelle & Li 2011) had shown empirically that TS dominates UCB on industrial datasets; this paper gives the theoretical scaffolding.

Why it matters this week. The proof structure — bound the probability that a sub-optimal arm's posterior dominates the optimal arm's posterior — generalises directly to Beta-TS (Agrawal & Goyal 2012), NeuralTS, and meta-Thompson. The frequentist bound $\tilde{O}(d^{3/2}\sqrt{T})$ is loose by a factor of $\sqrt{d}$ compared to LinUCB; Russo & Van Roy 2014 recovered the $d\sqrt{T}$ rate via an information-theoretic argument.

> "Despite its potential and empirical success, theoretical understanding of Thompson Sampling has been quite limited. ... We prove a high probability regret bound for Thompson Sampling for contextual bandits with linear payoff functions of $\tilde{O}(d^{3/2}\sqrt{T})$."

Commentary. The empirical-vs-theoretical gap is a recurring theme. Take the message: TS works, the theory is conservative, the production default is "LinTS with v tuned by replay on a held-out log".

4. Dudík, Langford, Li — *Doubly Robust Policy Evaluation and Learning* (ICML 2011, arXiv:1103.4601)

The DR estimator's modern home. Combines the IPS-style importance-weighted correction with a Direct-Method baseline, in exactly the form we derived in chapter 2.

Why it matters this week. DR is the OPE tool you should default to. Lower variance than IPS, lower bias than the Direct Method, and the formal double-robustness guarantee makes it the safest choice when you are unsure whether the propensities or the reward model are correct — and in practice you are always unsure.

> "The estimator is doubly robust: the estimator is consistent when either the reward model or the propensity model is correct. The doubly robust estimator typically improves upon the direct method and the IPS estimator since it uses both reward and propensity information."

Commentary. The same DR construction underlies the modern "self-normalised DR" (Thomas & Brunskill 2016), "MAGIC" for sequential RL (Thomas & Brunskill 2016), and the SWITCH-DR thresholding (Wang, Agarwal, Dudík 2017). One technique, three decades of papers.

5. Joachims, Swaminathan, de Rijke — *Deep Learning with Logged Bandit Feedback* (ICLR 2018; BanditNet, arXiv:1707.07724)

Turns the counterfactual risk minimisation (CRM) principle of Swaminathan & Joachims 2015 into a deep-net training loss. The key trick is the *equivariant* SNIPS objective with a Lagrangian rewrite that makes SGD stable on logged bandit data.

Why it matters this week. CRM + BanditNet is the only published recipe that makes "train a deep policy directly against logged data" work without falling into propensity overfitting. You will use the same loss in Week 12 to train the ATS re-ranker against logged shortlists.

> "[. . .] direct application of standard supervised learning techniques to logged bandit feedback is hindered by the bias inherent in propensity weighting and by the high variance of the IPS estimator. We address these issues by introducing a counterfactual risk minimisation principle and deriving a deep-network training objective."

Commentary. The Lagrangian rewrite to handle the self-normalisation divisor is non-obvious; read the appendix carefully. The same numerical recipe works for ranking (CLTR), recommendation, and ad allocation.

Bonus paper: Saito & Joachims — *Off-Policy Evaluation for Large Action Spaces via Embeddings* (ICML 2022, arXiv:2202.06317)

The most important applied OPE advance of the last five years. Standard IPS variance scales with the number of arms K; when actions admit a natural embedding $e(a) \in \mathbb{R}^{d_e}$ with $d_e \ll K$, the Marginalised IPS (MIPS) estimator replaces per-action propensities by marginal embedding propensities and cuts variance from $O(K/n)$ to $O(d_e/n)$.

Why it matters. synthetic insurance may have 6 brokers today but 600 next year; ATS has 10 000+ active requisitions at any time. Without MIPS, OPE on these scales is statistically void.

> "We show that the variance of the proposed estimator depends on the number of effective action embeddings, rather than the number of actions $|\mathcal{A}|$."

Commentary. The required assumption — conditional independence of the reward from a given (x, e) — is the embedding-sufficiency assumption. For broker matching, a good embedding is the broker's historical conversion profile by lead category; for jobs, the requisition skill vector. Practical: pick the most expressive embedding that still satisfies sufficiency by audit (residual check on a holdout).

Bonus paper: Bottou et al. — *Counterfactual Reasoning and Learning Systems* (JMLR 2013)

The Microsoft adCenter paper. 81 pages. Reads like a textbook on how to turn a real ad-auction system into a counterfactual-evaluation laboratory. Co-authored by Leon Bottou (SGD), John Langford (vowpal wabbit, contextual bandits at Microsoft), and the adCenter ML team.

Why it matters. This is the most honest published account of running OPE in production. It covers: how to randomise enough to log propensities without burning revenue; how to detect propensity overlap failures; how to compose multiple counterfactual estimators; how to write the dashboard a non-ML stakeholder will actually trust.

> "Causal inference is no harder than predictive analytics when the proper counterfactual data has been collected. Designing the data collection is the hard part."

Commentary. The line above should be tattooed on every applied ML engineer's wrist. The data collection (logging propensities, version hashes, sensitive attributes, delay outcomes) is 80% of the project; the estimator is 20%. Read this paper twice before touching a synthetic insurance / ATS production logger.

Bonus paper: Bietti, Agarwal, Langford — *A Contextual Bandit Bake-off* (JMLR 2021, arXiv:1802.04064)

A 47-page systematic empirical comparison of 21 contextual-bandit algorithms across 500+ datasets. The most useful paper if you actually have to pick an algorithm for production.

Why it matters. It collapses three decades of bandit research into a small number of practical recommendations. Spoiler: "cover with softmax exploration on top of boosted trees, with an offline DR-corrected loss" wins on the median problem; LinUCB / LinTS are competitive on linear-friendly datasets but dominated when reward is non-linear in the features.

> "We find that an algorithm that runs cover-greedy with an online softmax exploration policy, combined with a doubly-robust loss for offline optimisation, outperforms more sophisticated algorithms across the suite of datasets we consider."

Commentary. The lesson is not "cover always wins". The lesson is that the algorithm you pick matters less than (i) whether your features are good, (ii) whether you log propensities cleanly, and (iii) whether your offline evaluator is doubly robust. Defaults beat sophistication.

Glossary of terms used in this week's papers

Behaviour / logging policy (π_b). The policy that generated the offline log. You evaluate counterfactuals against π_b.

Target policy (π). The policy whose value $V(\pi)$ you want to estimate without deploying it.

Positivity / overlap. $\pi_b(a \mid x) > 0$ wherever $\pi(a \mid x) > 0$. Required for IPS / DR unbiasedness.

Importance ratio (w_i). $\pi(a_i \mid x_i)/\pi_b(a_i \mid x_i)$. Large weights cause variance; capping is the standard variance-control trick.

ESS. Effective sample size, $(\sum w_i)^2 / \sum w_i^2$. The size of the equivalent i.i.d. sample.

CRM. Counterfactual risk minimisation (Swaminathan & Joachims 2015): train a policy by minimising a self-normalised IPS objective with variance regularisation.

DR. Doubly robust: combine IPS-weighted residual correction with a direct-method baseline; unbiased if either component is correct.

MIPS. Marginalised IPS (Saito & Joachims 2022); use embedding propensities instead of action propensities.

Propensity overfitting. Re-fitting π_b from logs leaks reward into the propensity; biases IPS. Log π_b at decision time.

Delayed feedback. Reward arrives after a stochastic delay. Biases recent-policy evaluation downward. Mitigate by censoring or survival modelling (Chapelle 2014; Vernade et al. 2017).

Where to go next

- **Textbook**: Lattimore & Szepesvári, *Bandit Algorithms* (Cambridge UP, 2020) — chapters 7–10 cover UCB, KL-UCB, MOSS; 19–22 cover LinUCB and LinTS; 28–32 cover OPE and CRM.
- **Empirical state of the art**: Bietti, Agarwal, Langford, *A Contextual Bandit Bake-off* (JMLR 2021) — the right reference for which algorithm to use on real logs (spoiler: bag-of-trees with bootstrap-TS and an offline doubly-robust correction).

- **Large action spaces**: Saito & Joachims, *Off-Policy Evaluation for Large Action Spaces via Embeddings* (ICML 2022; MIPS). The variance-from-K-to-d_e trick is the most important practical advance of the past five years for matching applications.
- **Production**: Bottou et al., *Counterfactual Reasoning and Learning Systems* (JMLR 2013) — the Microsoft adCenter paper that spells out the full propensity-logging, OPE, and counterfactual-training pipeline. Mandatory reading before any production deployment.

Sourced from `research/bandits_matching_sota.md` sections 1–3 (stochastic, contextual, and OPE) and section 12 (production case studies).

Annotated reading order

If you have only one weekend, read in this order:

1. Bottou et al. 2013 (the production map). Skim sections 1–3, deep-read 4–6.
2. Auer, Cesa-Bianchi, Fischer 2002 (UCB1). Two pages of algorithm, 12 of proof.
3. Li et al. 2010 (LinUCB). Algorithm, replay evaluation, production results.
4. Dudík, Langford, Li 2011 (DR). Three pages of derivation.
5. Bietti, Agarwal, Langford 2021 (Bake-off). Skim the result tables, read sections 2 and 5.

If you have one focused day after the weekend, add:

6. Agrawal & Goyal 2013 (LinTS). The frequentist analysis, focus on Lemma 1.
7. Joachims, Swaminathan, de Rijke 2018 (BanditNet). The CRM Lagrangian.
8. Saito & Joachims 2022 (MIPS). The embedding-sufficiency assumption.

If you are about to deploy this in production, also read:

- Wu, Srikant et al. 2016 (Conservative Bandits).
- Chapelle 2014 (Delayed Feedback).
- Lattimore & Szepesvári 2020, chapter 32 (off-policy evaluation chapter).